

INQS 125 Game Theory in Popular Culture

Course Information

INQS 125 01
MWF 11:00-12:05
Graf 117

Contact Information

Instructor	Jennifer Nordstrom
Office	Taylor 205
Drop in Hours	MTWR 1:30-3:00 But feel free to drop by any time.
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Course Description

This course provides an investigation of game theory through film, television, and fiction. We will explore ideas such as perfect information, prisoner's dilemma, and volunteer's dilemma. We will examine the assumptions, mathematical models, and historical development of game theory. Students will learn the basic mathematical underpinnings of game theory and then apply these concepts to themes of conflict, cooperation, and trust.

Course Materials

The following is a list of primary resources for the course. There may be additional readings distributed during the course.

Required Texts

These text are available through the Linfield Bookstore.

- *Prisoner's Dilemma*, Poundstone, Doubleday, 1992.
- *They Say I Say*, 6th ed., Graff and Birkenstein, Norton, 2024.

Other Readings

These articles will be distributed in the course.

- “Battles of Wits and Matters of Trust: Game Theory in Popular Culture,” Nordstrom, *Mathematics and Popular Culture: Essays on Appearances in Film, Literature, Gaming, Television and Other Media*, eds, E. Sklar and J. Sklar, McFarland, 2012.
- “Game Theory and Literature,” Brams, *Games and Economic Behavior*, **6**, 32-54, 1994.
- “The Tale of Happiton,” *Metamagical Themas: Questing for the Essence of Mind and Pattern*, Hofstadter, Basic Books, 1985.

Required Films

These films are available to borrow in the Linfield Library. You may also watch them through your own services. If requested, I can arrange a group showing of these films.

- *Dr. Strangelove* (1964)
- *Return to Paradise* (1998)
- *The Dark Knight* (2008)
- *Crazy Rich Asians* (2018)
- *Footloose* (1984)

Other Required Materials

- Note cards for discussion questions. You can get these from me if you need some.
- Paper and pen or pencil. Bring to class every day. We will often do technology-free writing in class.
- Access to a printer for response papers and drafts. These need to be printed *before* class. There is access to free printing in the Taylor Hall Computer Lab (2nd floor), Graf (first floor), Library. Please give yourself sufficient time to deal with printing issues and travel time between buildings before papers are due.

INQS Learning Outcomes

The overarching goal of the Inquiry Seminar is to introduce students to the practices of inquiry, which form the foundation for the intellectual communities of the academy and the larger society. We believe this introduction is best accomplished by creating opportunities to conduct real inquiry within the classroom. We also recognize that the Inquiry Seminar is a beginning and that students will continue to develop and refine the skills and habits of inquiry across courses and disciplines during their four years of study. Specifically, the following list summarizes the learning outcomes for all Inquiry Seminars.

Objectives

1. Students frame key questions important to their own inquiry and to the understanding of a particular area of knowledge about which there is room for interpretation, ambiguity, and/or debate.
2. Students discuss, draft, compose, and reconsider answers to such questions in ways appropriate to the field and compelling to an intended audience.
3. Students engage and incorporate the voices of others to support their own learning and argumentation. In doing so, they will conduct research using library resources cited according to the ethical expectations of their academic community.
4. Students self-consciously and self-critically reflect on their own ways of thinking.

Topic Outline

The following is an outline of the course by main topics and major writing assignments.

The Assumptions

Topics include: perfect knowledge, rationality, value

Big Questions

- Do these assumptions really model society?
- What happens if we change the assumptions?
- What happens if we make assumptions that don't actually hold?

Readings and Films

- *The Prisoner's Dilemma*
- "Battles of Wits and Matters of Trust: Game Theory in Popular Culture"
- *Dr. Strangelove*

Question 1 Consider a game-theoretic scenario in a novel or film or your choice. How is the situation affected by the assumptions? Do all of the players seem to be using the same assumptions? How do the assumptions differ from standard game theory? What is the director or writer trying to convey about the characters through game theory? Your essay should be 3-5 pages. ▲

The Models

Topics include: zero-sum games, Chicken, Prisoner's Dilemma

Big Questions

- Are these reasonable models for conflict?
- How can we apply these models to political and societal situations?

Readings and Films

- *Return to Paradise*
- *The Dark Knight*
- *Crazy Rich Asians*
- *Footloose*

Question 2 *The Dark Knight* and *Return to Paradise* both use social dilemmas such as the Prisoner's Dilemma to comment on social issues such as trust and cooperation. Pick two characters from the films (can be the same film or different films) and relate their experience, choices, and motives to game theory. Some possible questions you can use to frame your essay:

- What game are they playing?
- What is the director trying to convey about the characters through the game?
- Is the director truly applying game theory?
- Is there a winner of the game?
- Did players choose optimal strategies?
- How would alternate strategies have changed the outcome of the game?

You are encouraged to develop your own question connecting the films and game theory. Be sure to provide evidence from the film(s) to support your claim. Your essay should be 3-5 pages. ▲

Group Dynamics

Topics include: multiperson Prisoner's Dilemma, Volunteer's Dilemma

Big Questions

- How can we model conflict and cooperation among groups?
- What are the societal impacts of cooperation and defection?

Readings and Films

- "Game Theory and Literature"
- "The Tale of Happiton"

Question 3 Research Paper. Choose a game-theoretic dilemma and examine how it has been applied in literature, fiction, television, and/or film. What does this dilemma convey about the characters, story, or society? Your paper should cite at least three academic sources *and* at least three novels, films, or television shows. Be sure to provide evidence to support your claim. You are encouraged to deviate from this basic frame with your own idea relating game theory to popular culture; however, you are required to get pre-approval for any paper idea that deviates significantly from the requirements. Your paper should be 8-10 pages. ▲

Use of Blackboard

This course uses [Blackboard \(Linfield Learn\)](#) for course management. You should use Blackboard for the following tasks

- Checking the daily schedule;
- Finding all assignments and due dates;
- Submitting response papers and essays;
- Getting feedback on your papers;
- Checking your grade during the course.

If you have questions about how to submit your work, see feedback, or find other information, please just ask. The schedule on Blackboard will have the most current information, including any adjustments to the schedule.

Some assignments are required to be turned in in hard-copy form. These include in class free-write assignments, draft for peer review, and peer review forms.

Discussion questions are to be turned in on note cards. You can get a note card from me, if you need one. Please get note cards before they are due.

Writing Center

When working on writing assignments for this class, I encourage you to take advantage of the Linfield Writing Center. The Writing Center is a drop-in service that provides help with all parts of the writing process—brainstorming, drafting, revising, refining. The Writing Center can help you with your thesis, organization, logic and reasoning, evidence, documentation, grammar and usage and more. The Writing Center is located in Nicholson Library. There is more information available on their website: [Linfield Writing Center](#).

The Writing Center offers in-person appointments. The hours are Monday-Thursday 12PM-9PM, and Friday 12PM-5PM in the Nicholson Library.

You can book appointments at [Writing Center Booking](#)

Grading Information

The following includes descriptions of the graded components of the course and the grading scale.

In-Class Writing

There will be daily in-class “free-write” questions. These are opportunities to practice putting your thoughts on paper. They will not be graded for content or grammar, but simply for turning them in. If you are not present for a free-write assignment, you can do the free-write before the next class. Free-writes must be in hard-copy form. These assignments will be scored 0 or 1 for submission.

- How to turn in: hard copy (generally hand-written) turned in during class.
- Late submission accepted? Yes. Before the next class.

Participation

You are expected to participate in class discussions and activities. You get the most out of this course if you regularly work with your peers. Part of your participation is coming to class prepared and on time. This means having your note card or response paper ready at the start of class. It also means making sure you have done the reading or film viewing before the discussion. Each class period will be scored 0 or 1 based on your attendance and preparation. These scores will be averaged with your in-class writing assignments.

Question Cards and Reading Responses

You will be required to turn in question cards and reading responses. Refer to the schedule in Blackboard for due dates. The cards should consist of a discussion question written on a 3x5 card; the reading response should be a one page, double-spaced, typed analysis of the day's reading or viewing. Both the card and response should reflect a thoughtful reading or viewing of the assignment. Both will be graded and must demonstrate critical thought and revision. Consequently, the cards and response papers will collectively have the weight of one essay. These assignments will *not* be accepted late, as they are an important part of the class discussion. If you are unable to attend class, you can email your question or response before class to be shared with your discussion group.

- How to turn in: submit response paper in Blackboard, cards turned in at the end of class.
- Late submission accepted? No.

Essays and Research Paper

During the term, you will be required to compose two analytic essays, one research paper, and one reflective essay. I will review any draft you bring to my office. I encourage you to seek help from me on your essays early and often. Topics for your essays will come from the in-class discussions, the assigned readings, and your own research. All essays must address a question-at-issue for the class. Submit electronically on Blackboard by the due date. You may request a no-questions-asked 24 hour extension as long as you contact me *before* the deadline.

- How to turn in: submit papers in Blackboard.
- Late submission accepted? Yes. A 24 hour extension is allowed if you request it before the deadline.

Peer Review

Before each major writing assignment, we will have a day of peer review. You are required to bring a hard-copy of your draft to be used for peer review. You will exchange papers with at least two other students in the class. You will be required to provide constructive feedback and responses to particular aspects of the papers you are reviewing. You will receive credit for having your draft and for providing feedback to your peers.

- How to turn in: Bring draft to the peer review class. Answer Peer Review questions for at least two other papers. Turn in the packet with your draft and peer review comments in class after you have submitted your final paper.

- Late submission accepted? Yes, however all peer review must be complete by paper submission deadline.

Final Presentation

You will be required to do a final presentation for the course. Your presentation will provide an 8-10 minute outline of the key ideas in your final research paper. You are required to attend all of the presentations. Your grade on your presentation will be reduced by 3 percentage points for each presentation you miss.

- How to turn in: submit slides on Blackboard.
- Late submission accepted? No. Due the at the end of the final exam time.

Grading System

Grades will be determined as follows:

In-class writing, peer review, participation, other assignments	10%
Question cards and reading responses	15%
Essay 1, Essay 2, Reflective Essay	45% (15% each)
Research Paper	20%
Final presentation	10%

Letter grades correspond to the following percentages:

A-, A	90-100%
B-, B, B+	80-89%
C-, C, C+	65-79%
D	55-64%

Important Due Dates

- [Question 1](#) due 11:59 pm Monday, Oct. 5
- [Question 2](#) due 11:59 Monday, November 2
- [Research Paper](#) due 11:59 pm, Friday, December 11
- [Final Presentation](#), Monday, December 7
- Final Presentations, Wednesday, December 9
- Final Presentations, Friday, December 11
- Reflective Essay due 10:30 am Friday, December 18

Final Exam Time

The final exam is scheduled for ***Friday, December 18 at 10:30 am.*** We will not be officially meeting during that time. However, all work for credit, including your presentation slides, your reflective essay, and any peer review packets must be turned in no later than the end of the final exam time, 12:30 pm.

Course Policies

Attendance:

You are expected to attend all classes. Discussion of the material cannot happen effectively without everyone being present and prepared. Your own writing is likely to suffer if you have not participated fully in all discussions. If you miss more than 3 classes without discussing your absences with me, you will no longer be able to make up missing in-class work. Similarly, if you are frequently late, you may lose the opportunity to make up missed work during that time.

Class Participation:

You are expected to participate daily in class discussions and activities. You need to come to class each day with a positive attitude and willingness to think about challenging ideas. You are expected to come prepared to class by having done the required readings and assignments. Our class should function as an inclusive community. Everyone needs to make room at the table for a diversity ideas and experiences. Disagreement is inevitable, and in fact, often the goal. But I expect everyone to treat each other with respect.

Late Work:

See specifics in [Grading Information](#). Work that can be accepted late: in-class work by the next class period; Essay 1, Essay 2, Research Paper within 24 hours if requested; peer-review by the paper due date.

Cell Phones:

Cell phones must be off and put away during class. Laptop computers and tablets are not required in class. If you need to use one for classroom purposes, please use them in ways that are focused on the course and the activities of the class.

Electronic Recording/ Content Sharing:

I may opt to record the classroom activities for instructional purposes and post them to the cloud. The electronic recording of classroom lectures, discussions, simulations, and other course-related activity is governed by Linfield's Classroom Recording Policy (Faculty Handbook, VII.26 and Student Policy Guide). Students do not have permission to record any Zoom meetings. Students do not have permission to distribute or share any recorded content from Zoom meetings.

Use of AI in Student-Generated Work:

Students in this course are expected to use AI tools, such as Chat GPT, Gemini, Claude, and Copilot in an ethical and responsible manner. Since one of the purposes of this course is to learn how to effectively generate and communicate your own ideas, we will discuss how AI can help or hinder learning. Use of AI tools, like any other academic work that is not entirely the student's own, must be cited. Work for a grade that is not primarily in the student's own words and properly cited, will be considered plagiarized.

If a violation is suspected, the first action will be a conversation about why your work may be read as AI generated. Work that is deemed not the student's own after this conversation will receive a 0 on the entire assignment/ paper/ presentation.

Some AI uses that are appropriate: finding sources for a paper, general brainstorming, helping you create an outline for a paper based on your own ideas, helping summarize key concepts with additional examples, helping create a study plan for balancing your workload. Uses that are not appropriate: generating the body of a paper, generating sources that you do not actually incorporate into a paper, having AI write a paper based on your outline.

In general, papers must incorporate YOUR ideas and reflect YOUR understanding of the concepts and issues. Generic papers (whether or not they were written by AI) will not be able to earn full credit.

Just as I expect you to avoid the use of AI in your writing, I will avoid the use of AI in my grading of your work. This means both in evaluating your work and in assigning grades. I may use AI detection technology, but any determination of inappropriate AI use will be based primarily on other evidence, such as lack of individual style, changes in tone, inconsistency with the course, and your overall development as a writer. I will always be clear when I have used AI, as I expect the same from you.

Linfield Policies

Disability Statement:

Students with disabilities are protected by the Americans with Disabilities Act and Section 504 of the Rehabilitation Act. If you are a student with a disability and feel you may require academic accommodations please contact Learning Support Services (LSS), as early as possible to request accommodation for your disability. The timeliness of your request will allow LSS to promptly arrange the details of your support. LSS is located in Melrose Hall 020 (503-883-2562), or LSS@linfield.edu. We also encourage students to communicate with faculty about their accommodations.

Academic Integrity Policy:

Linfield University operates under the assumption that all students are honest and ethical in the way they conduct their personal and scholastic lives. Academic work is evaluated on the assumption that the work presented is the student's own, unless designated otherwise. Anything less is unacceptable and is considered a violation of academic integrity. Furthermore, a breach of academic integrity will have concrete consequences that may include failing a particular course or even dismissal from the university. Violations of academic integrity include but are not limited to the following:

Cheating: Using or attempting to use unauthorized sources, materials, information, or study aids in any submitted academic work; changing answers after graded work has been returned; making unauthorized changes to an exam, quiz, or assignment.

Plagiarism: Submission of academic work that includes material copied or paraphrased from published or unpublished sources without proper documentation. This includes self-plagiarism, the submission of work created by the student for another class unless they receive consent from both instructors.

Fabrication: Deliberate falsification or invention of any information, data, or citation in academic work.

Facilitating Academic Dishonesty: Knowingly helping or attempting to help another to violate the university's policy on academic integrity.

For specific information about the use of AI, see [Use of AI in Student-Generated Work](#).

Any form of academic dishonesty will result in a 0 on that assignment/quiz/ exam. Additionally, academic dishonesty may result in a failing grade in the course. See the [Linfield Academic Integrity Policy](#) in the Linfield Catalog for information on the procedure to be used in dealing with academic dishonesty.

Sexual Misconduct and Relationship Violence and Title IX:

Linfield University faculty are committed to supporting students and fostering a campus environment free of sexual misconduct and relationship violence. If a student chooses to disclose to a faculty or staff member an experience related to sexual misconduct, sexual assault, domestic violence, dating violence, or stalking, all faculty and staff are obligated to report this disclosure to the Linfield Title IX Coordinator by emailing titleix@linfield.edu. Upon receipt of the report, the Title IX Coordinator will contact you to inform you of your rights and options and connect you with support services. If you would rather share information about these experiences with an employee who does not have these reporting responsibilities and can keep the information confidential, please visit confidential resources: linfield.edu/safety/sexual-misconduct/reporting-options/confidential.html.

For more information about your rights and reporting options at Linfield, including confidential reporting options, please visit Linfield's [Know Your Options Page](#) on the University website. Support services are offered to all Linfield students regardless of whether or not they report. Still have questions? Email titleix@linfield.edu.

Commitment to Diversity and Inclusion:

Linfield University honors human rights and academic freedom, celebrates diverse cultures, fosters a climate of mutual respect, and promotes an inclusive environment that affirms the value of all persons. Dimensions of diversity can include sex, race, age, national origin, immigration status, ethnicity, gender identity and expression, intellectual and physical ability, sexual orientation, income, faith and non-faith perspectives, socio-economic class, political ideology, education, primary language, family status, military experience, cognitive style, and communication style. In a multi-perspective intellectual space, challenges to our beliefs and ideas are part of the learning process and can provide opportunities for growth. Reasoning, thoughtfulness, and open dialogues that honor the dignity of everyone is expected.